

HINTS AND SOLUTIONS

Q.1 A

Sol. $I_{\max} = (\sqrt{I_1} + \sqrt{I_2})^2 = \frac{9}{4} I_1$
 $\Rightarrow S_2P - S_1P = n\lambda$ for maxima
 $\sqrt{d^2 + x^2} - x = n\lambda$
 $x = \frac{d^2 - n^2\lambda^2}{2n\lambda}$

Q.2 A

Sol. Path difference = $\frac{\lambda}{2}(2n-1)$ for constructive interference due phase change at reflection.

Q.3 D

Sol. $2\mu t = \left(n + \frac{1}{2}\right)\lambda$
 for $\lambda = 4000\text{\AA}$, $n = 7$
 and for $\lambda = 7000\text{\AA}$, $n = 3.78$
 There are four integers between the two values of n .

Q.4 D

Sol. $x = \frac{(\mu - 1)t\beta}{\lambda}$
 $2\beta = \frac{(1.4 - 1)t\beta}{5000 \times 10^{-10}}$
 $t = 25 \times 10^{-7} = 2.5 \mu\text{m}$

Q.5 C

Sol. The intensity at P will depend on the path difference and also the initial phase difference between the two interfering light sources. If at point P, the interfering waves has a path difference of $2\pi/3$ then the resultant intensity at P = I.

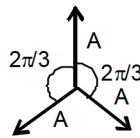
Q.6 A

Q.7 C

Sol. $\phi = \frac{2\pi}{\lambda} \times \frac{\lambda}{2} = \pi$
 $I_A = I_0 \cos^2\left(\frac{\pi}{2}\right) = 0 \quad \dots(1)$
 $I_B = I_0 \cos^2\left(\frac{\pi}{4}\right) = \frac{I_0}{2} \quad \dots(2)$

Q.8 D

Sol. $\vec{A}_r = \vec{A}_1 + \vec{A}_2 + \vec{A}_3 = 0$
 $\Rightarrow I_r = 0$



Q.9 B

Sol. $\frac{I_0}{2} = I_0 \cos^2 \frac{\Delta\phi}{2}$

$$I_0 \cos^2 \frac{\Delta\phi}{2} = \frac{1}{2} \Rightarrow \Delta\phi = \frac{\pi}{2}$$

$$\Delta x = \lambda/4 = (\mu - 1)t$$

$$t = \lambda/2$$

Q.10 A

Sol. $(\mu - 1)t = n\beta$

$$\frac{(\mu_1 - 1) \times 1.8 \times 10^{-5}}{(\mu_2 - 1) \times 3.6 \times 10^{-5}} = \frac{18\beta}{9\beta}$$

$$(\mu_1 - 1) = 4(\mu_2 - 1)$$

$$4\mu_2 - \mu_1 = 3$$

Q.11 AC

Q.12 ABC

Sol. (A) as biprism moves backward its distance from S decreases, hence distance between S_1 & S_2 decreases

$$\therefore d \text{ decreases in } \beta = \frac{\lambda D}{d} \Rightarrow \beta \text{ increases.}$$

(B) As screen is moved away D increases $\therefore \beta$ increases

(C) As α decreases as $S_1S = S_2S = (\mu - 1)A \times SP$ decreases
 α decreases $\therefore \beta$ increases

(D) Slit width affects intensity & not width of fringe

Q.13 ABCD

Sol. When $D \gg d$, rays from S_1 and S_2 are nearly parallel.

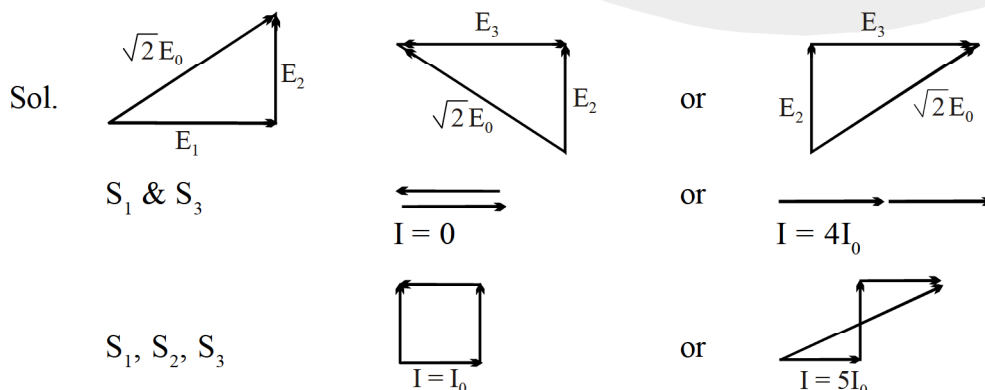
Which gives $\Delta x = d \sin \theta$

Also, $\Delta x = d \sin \theta = n\lambda$, $\sin \theta = n\lambda$.

Equally spaced fringes occur for small θ i.e. $d \gg \lambda$.

If $d = 3\lambda$, the possible path difference on the screen are $\pm 2\lambda, \pm \lambda, 0$.

Q.14 BC

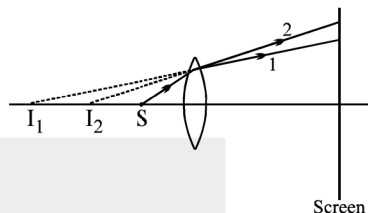


Q.15 C

Q.16 [Ans. (A) P,Q; (B) P,Q,R,S; (C) P]

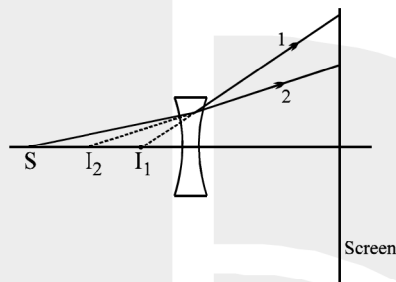
[Sol.(A)] For convex lens

$|f| \uparrow$ Converging power reduces, virtual image is formed closer to lens hence closer to screen



For concave lens

$|f| \uparrow$ diverging power $\downarrow$



I moves away

Hence (P), (Q)

(B) $|u| < |f| \Rightarrow$ virtual image in all cases

(P), (Q), (R), (S)

(C) Say $u = -u_0$

(i) converging lens

$$m = \frac{f}{f + (-u_0)} = \frac{f}{f - u_0}$$

if $u \downarrow$ $m \downarrow$ thus distance between images decreases

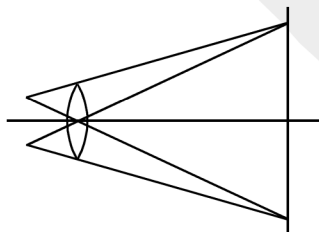
(ii) diverging lens

$$m = \frac{-f_0}{-f_0 + (-u_0)} = \frac{f_0}{f_0 + u_0}$$

if $u_0 \downarrow$
 $m \uparrow$

thus distance between images increases

(D)



In rest of cases no common region

Q.17 [Ans. (a) $\frac{2\pi}{\lambda} \left(\frac{yd}{D} \mu_2 - t(\mu_1 - \mu_2) \right)$; (b) 2.25 mm; (c) -0.3 mm]

Q.18 [Ans. 0.16 m]

Sol. $PR = \frac{h}{\sin \theta}$; $QR = PR \cos 2\theta = \frac{h \cos 2\theta}{\sin \theta}$

$$\Delta X = PR - RQ + \frac{h}{2}$$

$$\Delta X = \frac{h}{\sin \theta} (1 - \cos 2\theta) + \frac{h}{2} = 2h \sin \theta + \frac{h}{2}$$

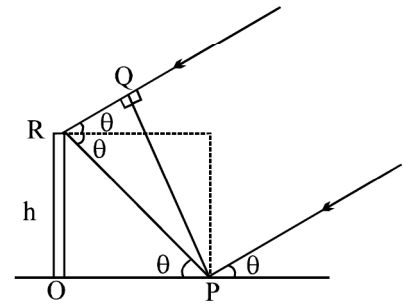
$$\Delta X \cong 2h\theta + \frac{h}{2}$$

$$\sin \theta = \theta$$

Maximum to maximum

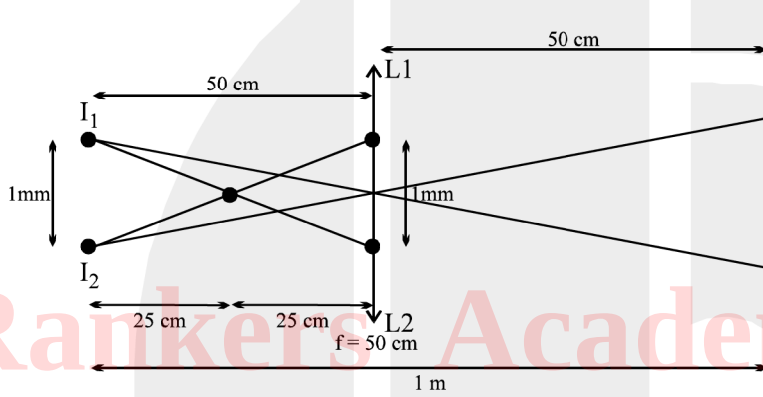
$$\lambda = 2h(\theta_2 - \theta_1) + 0$$

$$\lambda = 2(4)[0.03 - 0.01] = 2(4)(0.02) = 0.16 \text{ m}$$



Q.19 [Ans 0.6 mm]

Sol._{nj}



$$u = -25$$

$$\frac{1}{v} - \frac{1}{-25} = \frac{1}{50} \Rightarrow \frac{1}{v} = -\frac{1}{50} \Rightarrow v = -50$$

$$\beta = \frac{6 \times 10^{-7} \times 1}{10^{-3}} = 6 \times 10^{-4} = 0.6 \text{ mm}$$